Finite critical estimates and supplied-cube partitions

Companion proof and reproducibility record

The Eriksson Programme

Preparing assistant: OpenAI Codex; not an independent reviewer

5 November 2026

Abstract

This companion collects two finite classical derivations and their exact reproduction record within the existing coarse-information dossier. It supplies the critical rectangle Sobolev estimate and a linear-size partition for supplied disjoint lattice cubes, including discrete boundary ownership and the actual fixed-penalty energy adapter. It retains the coefficient-scale assumptions and does not derive suitable cubes from a critical inverse moment. The unrestricted rectangular question remains open. The document is an unreviewed companion, not a new AIRR paper, acceptance, peer review or an interacting or continuum Yang–Mills result.

Reading convention and provenance

The two parts below have separate scopes. Part one is a local inequality; it is not an algorithm producing a globally adequate partition. Part two constructs a partition only when actual disjoint cubes and the stated coefficient bounds have already been supplied. Their conclusions must not be composed to discard those assumptions.

The dimension symbol D is common to both parts. The constants named A_D are *local to their respective parts*: the Sobolev constant and the geometric partition constant are different. Each is explicitly defined at its point of use; neither is substituted for the other. All lattice norms use counting measure. Internal axis edges are counted once; periodic wrap edges may only be omitted using nonnegativity.

The classical entropy/projection/coarea derivation and de Berg's compressed splitting method are credited in the respective parts. We claim no new general Sobolev or BSP principle, original priority or independent discovery. This companion consolidates source fragments from units137–140, with source and existing evidence pinned in the accompanying manifest. It does not amend the frozen reviewed PDF or inherit its model assessments.

Contents

1 A finite proof of the critical rectangle estimate **3**
1.1 Finite projection inequality, with proof 3
1.2 Relative isoperimetry on a finite cube 3
1.3 Endpoint counting-norm Sobolev–Poincare inequality 4
1.4 Reflection with all aspect-ratio multiplicities 4
1.5 Positive original edge weights and field coordinates 5
1.6 Necessity of geometry and status 5

2 A finite rectangular partition of disjoint lattice cubes **6**
2.1 Proper-intersection corner bound 6
2.2 A multiple-object stress family 6
2.3 Constructing linearly many vertex-free rectangles 7
2.4 A uniform occupancy bound for the terminal rectangles 7
2.5 From open geometry to an actual integer partition 8
2.6 The actual coefficient and fixed-penalty consequence 8

3 Finite reproduction evidence and its limits **10**

4 Model assessments and review context **10**

5 Unresolved obligations and reproducibility instructions **11**

1 A finite proof of the critical rectangle estimate

This supplement derives the balanced-cube endpoint used by the rectangle adapter, rather than importing a local Sobolev estimate as an external input. The method is classical: finite entropy proves the projection inequality, which yields relative edge isoperimetry, layer sets, and a power argument. We claim neither a new general Sobolev principle nor a solution of the open critical rectangular-partition problem. This is part of the existing dossier.

Fix $D \geq 3$, and put

$$p = D/2, \quad q = \frac{2D}{D+2}, \quad q' = \frac{2D}{D-2},$$
$$\alpha = \frac{2(D-1)}{D}, \quad c_D = D(1-2^{-1/D}), \quad K_D = \frac{2\alpha D^{1/q'}}{c_D}.$$

An internal axis edge is an unordered adjacent pair, counted once. For its positive-axis orientation $e = (x, x+e_i)$, write $du(e) = u(x+e_i) - u(x)$.

1.1 Finite projection inequality, with proof

For a nonempty finite $A \subset \prod_{j=1}^{D} X_j$, let π_i delete coordinate i. Then

$$|A|^{D-1} \leq \prod_{i=1}^{D} |\pi_i A|. \tag{1}$$

Here is a finite proof, including the entropy facts needed for this use. For a finite probability vector r, define $H(r) = -\sum r \log r$ with $0 \log 0 = 0$. Concavity follows from $(-t\log t)'' = -1/t < 0$ and continuity at zero. Consequently mixing conditional distributions cannot decrease their average entropy: $H(Y \mid Z) \geq H(Y \mid Z, W)$. Factoring finite joint probabilities proves the chain rule $H(X) = \sum_j H(X_j \mid X_1, \ldots, X_{j-1})$. Also $H(Y) \leq \log |\operatorname{supp} Y|$: Jensen's inequality for log gives $\sum_y r_y \log(1/r_y) \leq \log \sum_y r_y / r_y$ on the positive support. For X uniform on A, $H(X) = \log |A|$, while

$$\begin{aligned}\sum_i H(X_{-i}) &= \sum_i \sum_{j \neq i} H(X_j \mid X_{<j} \setminus X_i) \\ &\geq \sum_i \sum_{j \neq i} H(X_j \mid X_{<j}) = (D-1)H(X).\end{aligned}$$

The notation in the conditioning set means omit X_i only when $i < j$. Bounding each $H(X_{-i})$ by $\log |\pi_i A|$ and exponentiating proves (1). No projection inequality is an unproved hypothesis in the sequel.

1.2 Relative isoperimetry on a finite cube

Let $Q_s = \{0, \ldots, s-1\}^D$, $s \geq 2$, and let $\partial_E A$ consist of internal cube edges crossing A and its complement. If $0 < |A| \leq |Q_s|/2$, then

$$|\partial_E A| \geq c_D |A|^{(D-1)/D}. \tag{2}$$

Indeed, put $a = |A|$, $P_i = |\pi_i A|$. At most a/s occupied fibers in direction i are full. Every occupied nonfull fiber contains an internal crossing edge, because the fiber is a connected interval. Edges in distinct fibers/directions are different. Thus projection inequality and AM–GM give

$$|\partial_E A| \geq \sum_i (P_i - a/s) \geq D a^{(D-1)/D} - Da/s \geq D(1-2^{-1/D}) a^{(D-1)/D}.$$

Empty A satisfies the same inequality trivially. The boundary is the EDGE boundary, not the vertex boundary in a separate graph theorem.

1.3 Endpoint counting-norm Sobolev–Poincare inequality

For every real u on Q_s, including $s = 1$,

$$\sum_{Q_s} |u - \bar{u}_{Q_s}|^2 \le K_D^2 \left(\sum_{e \subset Q_s} |du(e)|^q \right)^{2/q}. \tag{3}$$

For $s = 1$ both sides vanish. For a nonnegative v supported on at most half the cube, layer-set decomposition and the triangle inequality in $\ell^{D/(D-1)}$, followed by (2), imply

$$\|v\|_{D/(D-1)} \le \int_0^\infty |\{v > t\}|^{(D-1)/D}\, dt \le c_D^{-1} \int_0^\infty |\partial_E \{v > t\}|\, dt = c_D^{-1} \sum_e |dv(e)|.$$

The integrals are finite sums of step lengths between values of v; the last equality follows edgewise from the length of the interval between its two endpoint values. Apply this inequality to v^α. For $a, b \ge 0$, the fundamental theorem of calculus gives

$$|a^\alpha - b^\alpha| \le \alpha |a - b| (a^{\alpha-1} + b^{\alpha-1}).$$

Since $\alpha D/(D-1) = 2$ and $(\alpha - 1)q' = 2$, Holder and the degree bound $\deg_{Q_s}(x) \le 2D$ give

$$\begin{aligned} \|v\|_2^\alpha &\le \frac{\alpha}{c_D} \Big(\sum_e |dv(e)|^q \Big)^{1/q} \Big(\sum_{e=\{x,y\}} (v(x)^{\alpha-1} + v(y)^{\alpha-1})^{q'} \Big)^{1/q'} \\ &\le \frac{2\alpha D^{1/q'}}{c_D} \Big(\sum_e |dv(e)|^q \Big)^{1/q} \|v\|_2^{\alpha-1}. \end{aligned}$$

The second inequality uses $(a+b)^{q'} \le 2^{q'-1}(a^{q'} + b^{q'})$ and $\sum_e (v(x)^2 + v(y)^2) \le 2D\|v\|_2^2$. If $v = 0$ the conclusion is immediate; otherwise division proves $\|v\|_2 \le K_D \|dv\|_q$.

Choose a counting median m of u and set $v_+ = (u - m)_+$, $v_- = (m - u)_+$. Both supports have size at most half the cube. For each edge,

$$|dv_+(e)|^q + |dv_-(e)|^q \le |du(e)|^q.$$

For endpoint values on the same side of m this is equality in the nonzero part; on opposite sides it is $a^q + b^q \le (a+b)^q$. Since $2/q \ge 1$, addition of the two squared estimates gives

$$\|u - \bar{u}_{Q_s}\|_2^2 \le \|u - m\|_2^2 \le K_D^2 \big(\|dv_+\|_q^q + \|dv_-\|_q^q \big)^{2/q} \le K_D^2 \|du\|_q^2.$$

This proves (3) without an external graph, ball-adapter, coefficient, or boundary-condition hypothesis.

1.4 Reflection with all aspect-ratio multiplicities

Let $R = \prod_i \{0, \dots, l_i - 1\}$, $V = \prod_i l_i$, $L = \max_i l_i$, and $\kappa = L^2/V^{2/D}$. If $V = 1$ the desired variance bound is zero. Otherwise $L \ge 2$. Put $s = 2L+1$, $r = s^D/V$ and fold Q_s coordinatewise onto R by the endpoint-repeating pattern of period $2l_i$,

$$0, 1, \dots, l_i - 1, l_i - 1, \dots, 1, 0, 0, 1, \dots.$$

For $l_i = 1$ the fold is constant. Since $s/(2l_i) > 1$, each original coordinate value has at least $2\lfloor s/(2l_i) \rfloor \ge s/(2l_i)$ preimages and at most $s/l_i + 2 \le 3s/l_i$ preimages. An original internal edge has at most the same upper count in its own direction; each full period traverses it twice.

Adjacent folded values either coincide or are an original internal edge, possibly with reversed orientation. Thus for the unweighted extension $\widetilde{u}$,

$$\mathrm{Var}_{Q_s}(\widetilde{u}) \geq 2^{-D} r \,\mathrm{Var}_R(u), \qquad \sum_{e\subset Q_s} |d\widetilde{u}(e)|^q \leq 3^D r \sum_{e\subset R} |du(e)|^q.$$

The variance inequality holds by applying the preimage lower bound at the mean of $\widetilde{u}$, and then minimizing over constants on R. Apply (3). Since $2/q - 1 = 2/D$ and $s \leq 3L$,

$$\mathrm{Var}_R(u) \leq A_D \kappa \left(\sum_{e\subset R} |du(e)|^q \right)^{2/q}, \qquad A_D = K_D^2 2^D 3^{2D/q+2}. \tag{4}$$

The factor r is not bounded independently of aspect ratio; it was kept until it became $s^2/V^{2/D}$. Singleton coordinate sides cause no problem.

1.5 Positive original edge weights and field coordinates

For arbitrary positive ORIGINAL edge coefficients b_e, Holder on R with exponents $2/q$ and $2/(2-q)$ gives

$$\left(\sum_e |du|^q \right)^{2/q} \leq \Big(\sum_e b_e |du|^2 \Big) \Big(\sum_e b_e^{-q/(2-q)} \Big)^{(2-q)/q}.$$

Here $q/(2-q) = D/2 = p$. Consequently

$$\mathrm{Var}_R(u) \leq A_D \kappa \Big(\sum_{e\subset R} b_e^{-D/2} \Big)^{2/D} \sum_{e\subset R} b_e |du(e)|^2. \tag{5}$$

No reflected weights were introduced: reflection was completed before this Holder step. For $b_{(x,x+e_i)} = a(x) > 0$, each vertex supplies at most D internal starting edges. Thus the inverse edge mass is at most $D \sum_R a^{-D/2}$, proving exactly the scalar-start estimate

$$\mathrm{Var}_R(u) \leq A_D D^{2/D} \kappa \Big(\sum_R a(x)^{-D/2} \Big)^{2/D} \mathcal{E}_{a,R}(u).$$

For any fixed finite real internal/field dimension, apply this scalar inequality to each coordinate and sum; its coefficient factor is common. There is no additional internal-dimension factor and no replacement of original edges by enlarged-region energy.

1.6 Necessity of geometry and status

For uniform $a > 0$ and a longest coordinate of length $L \geq 2$, put $u(x) = x_i - (L-1)/2$. Direct sums give

$$\mathrm{Var}_R(u) = V(L^2-1)/12, \qquad \mathcal{E}_{a,R}(u) = aV(L-1)/L,$$

so their ratio is $L(L+1)/(12a) \geq L^2/(12a)$. Since $\kappa(\sum_R a^{-D/2})^{2/D} = L^2/a$, the geometric dependence cannot be uniformly dropped even for uniform weights. This is not a global counterexample to a partition algorithm or to the physical divergence-zero observation problem.

For attribution and comparison see S. Andres, J.-D. Deuschel and M. Slowik, *Harnack inequalities on weighted graphs and some applications to the random conductance model*, arXiv:1312.5473v6, Section 2.2, (2.5)–(2.8). Their local normalized inequalities are classical prior context; this proof does not import their theorem, graph assumptions or Harnack conclusions. The finite projection/entropy, coarea and power steps above are classical, not a new general method. The gain is a complete finite derivation of the specific input previously left external. No Lean execution, independent review, original-priority claim, new physical hypothesis removal or new AIRR paper follows from this supplement. The frozen PDF candidate and its actual reports/permissions remain unchanged; integration would create a new exact artifact and must not inherit their assessments.

2 A finite rectangular partition of disjoint lattice cubes

The linear partition for supplied disjoint cubes can be derived directly, without importing a general fat-object or low-density theorem. We use the classical compressed cube-splitting strategy of de Berg, UU-CS-1998-12, Section 2.1 and Lemma 2.1, and prove the required special-case geometry and all discrete/energy adapters below. This is not an original general BSP method. No time-complexity, measured implementation or Lean claim is made.

Fix $D \geq 3$, an integer $n \geq 1$, and pairwise disjoint nonempty integer cubes $S_k = \prod_i \{b_{ki}, \ldots, b_{ki} + L_k - 1\}$ in the common root $X = \{0, \ldots, n-1\}^D$, for $1 \leq k \leq J$. Set

$$Q_k = \prod_i [b_{ki} - 1/4, b_{ki} + L_k - 3/4].$$

These are positive-side closed solid cubes, including $L_k = 1$. They are strictly disjoint: disjoint integer intervals in a separating coordinate leave a real gap at least $1/2$. Their integer points are exactly S_k, and each such point is interior to Q_k. All corners of the Q_k lie strictly inside $[-1, n]^D$.

2.1 Proper-intersection corner bound

Let C be any axis-aligned cube of side $s > 0$ with no corner of any Q_k in its interior. Then at most 2^D of the Q_k meet $\operatorname{int} C$. Indeed, if Q_k of side $a < s$ properly meets C, each of its coordinate intervals has an endpoint in the interior of the corresponding interval of C. Choosing these endpoints gives a corner of Q_k in $\operatorname{int} C$, a contradiction. Thus every properly meeting Q_k has $a \geq s$. In each coordinate its interval then contains at least one endpoint of the interval of C; otherwise it would lie strictly inside that shorter interval. Hence Q_k contains a corner of C. Strict disjointness makes these corner choices injective, proving the bound.

The qualifier "properly" is essential. Closed boundary-only contact must not be counted here. For $C = [0, 1]^D$ and any integer $K \geq 1$, put $\epsilon = 1/(4K + 4)$ and

$$Q_j = [-\epsilon, 0] \times [(2j+1)\epsilon, (2j+2)\epsilon] \times [\epsilon, 2\epsilon]^{D-2}, \qquad 0 \leq j < K.$$

These K strictly disjoint small cubes touch C but none meets its interior or has a corner in its interior. Thus the corresponding closed-contact bound would be false. This boundary guard does not assert that de Berg's theorem is false; our construction below explicitly uses open-cell intersections and interior lattice points.

2.2 A multiple-object stress family

The corner-free property does not imply that a query cube meets only one input cube. This can fail even for the quarter-offset solids of equal-side integer cubes, not merely for arbitrary real objects. For $D \geq 2$, put $n = 4$, $C = [1, 2]^D$, and, for $\sigma \in \{0, 1\}^{D-1}$, set

$$b_\sigma = (2\sigma_1, \ldots, 2\sigma_{D-1}, 1), \quad L_\sigma = 2, \qquad S_\sigma = \prod_{i=1}^{D} \{b_{\sigma i}, b_{\sigma i} + 1\}.$$

The 2^{D-1} integer cubes are pairwise disjoint: distinct binary words have a coordinate with the disjoint integer intervals $\{0, 1\}$ and $\{2, 3\}$. Their solids have, in that coordinate, intervals $[-1/4, 5/4]$ and $[7/4, 13/4]$, with a gap $1/2$. In the last coordinate every solid has interval $[3/4, 9/4]$. Consequently every solid properly meets C, while no solid corner is interior to C: every such corner has last coordinate $3/4$ or $9/4$. Thus a universal proper-occupancy bound for corner-free query cubes in this exact lattice class must be at least 2^{D-1}. This neither proves sharpness of 2^D nor asserts that this particular C occurs as a leaf of our root-based recursion. It rules out replacing the occupancy argument by a one-object assumption. The finite unit139 baseline leaves all had occupancy at most one, so those runs did not test this phenomenon.

2.3 Constructing linearly many vertex-free rectangles

Let V be the set of all corners of the Q_k, so $N = |V| = 2^D J$. For $J = 0$ use the single root rectangle and skip this construction. For $J > 0$ begin with the cube $[-1, n]^D$. At a current cube C keep exactly $V_C = V \cap \operatorname{int} C$. If V_C is empty, stop at that cube. Otherwise consider its 2^D open half-side children.

If the points are not all in one open child, perform all midpoint cuts and recurse on the nonempty child point sets. Points on the cuts are removed from subsequent interior sets. If all points lie in one open child, reflect and translate coordinates so the common corner is zero and $C = [0, s]^D$. Put

$$a = \max\{x_i : x \in V_C,\ 1 \le i \le D\}, \qquad C' = [0, a]^D.$$

Then $0 < a < s/2$. Successively cut at the D upper facet planes of C', producing C' and the D boxes

$$E_i = [0, a]^{i-1} \times [a, s] \times [0, s]^{D-i}, \qquad 1 \le i \le D.$$

Use disjoint boundary ownership for these boxes. Only C' can contain points of V in its interior, and at least one point of V_C lies on its boundary. Stop at the other boxes and recurse inside C' if needed. Every intermediate or terminal cell is full-dimensional; a boundary ownership convention never creates a separate hyperplane cell.

Here is a direct finite termination and size proof. Among active cells, let $n_C = |V_C| > 0$ and use the nonnegative integer potential

$$\Psi = \sum_{C \text{ active}} (2n_C - 1).$$

Initially $\Psi = 2N - 1$. If a processed cell loses r points to new cut boundaries and has t nonempty child point sets, the decrease is $2r + t - 1$. In the shrinking step $r \ge 1$. In a midpoint step either $r \ge 1$ or $t \ge 2$. Thus every event decreases Ψ by at least one, so there are at most $2N - 1$ events. Each event creates at most $2^D - 1$ additional leaves, since a shrinking step creates only D extra leaves. The number M of terminal rectangles satisfies

$$M \le 1 + (2N - 1)(2^D - 1) \le 2^{D+1}(2^D - 1)J. \tag{6}$$

Tracking exactly $V \cap \operatorname{int} C$ ensures that every terminal rectangle has no corner of an input cube in its interior. This argument is a construction and a decreasing integer potential, not a supplied partition certificate. It does not prove a fast implementation bound.

2.4 A uniform occupancy bound for the terminal rectangles

Terminal cubes meet at most 2^D input cubes properly, by the corner bound. A terminal noncube has the form E_i above after a coordinate reflection. It is contained in the slab $[0, s]^{i-1} \times [a, s] \times [0, s]^{D-i}$. Cover this slab by 2^{D-1} closed cubes of side $s - a$: in coordinate i use $[a, s]$, and in every other coordinate choose either $[0, s - a]$ or $[a, s]$. These intervals cover $[0, s]$ since $a < s/2$. Every covering cube lies inside the parent cube and outside the interior of C', and its interior contains no point of V. The corner bound therefore applies separately to each covering cube.

If an input cube properly meets E_i, the intersection contains an open set. A point of that set avoiding the finitely many covering-facet planes belongs to the interior of a covering cube. Thus all proper intersections are counted by the cover, even if some particular point lies on a cover boundary. Consequently each terminal rectangle properly meets at most

$$T_D = 2^{D-1} 2^D = 2^{2D-1} \tag{7}$$

input cubes. The covering argument, including its boundary qualification, is proved here; no arbitrary-box unclutteredness is assumed.

2.5 From open geometry to an actual integer partition

For each terminal rectangle list the at most T_D cubes meeting its interior. Refine by all their facet coordinates. There are at most $2T_D$ cut coordinates per axis, so at most $(2T_D+1)^D$ refined rectangles per terminal cell. All cuts are axis-parallel. Now assign every splitting hyperplane to one child consistently, intersect with X, and delete empty integer sets. The result is a disjoint cover of X by integer rectangles. Taking integer points and deleting empty sets does not increase the count. More explicitly, use the uniform convention $[l_i, h_i)$ in every axis of every cell. A coordinate split at t then assigns the plane to the upper child. The two intervals $[l,t)$ and $[t,h)$ partition $[l,h)$ exactly, even when t is an integer. By induction through every cut, these half-open cells partition $[-1,n)^D$; intersecting with X gives

$$R = \prod_i (\{\lceil l_i \rceil, \dots, \lceil h_i \rceil - 1\} \cap \{0, \dots, n-1\}).$$

This formula is also valid for negative or integer endpoints. An empty axis makes the entire integer cell empty; no lattice point is assigned twice or lost when those cells are deleted. The outer upper face at n contains no point of X.

To check membership rigorously, suppose an assigned integer point x lies in S_k. It is interior to Q_k, while its assigned terminal rectangle has x in its closure. A small ball about x lies in Q_k and meets the interior of that full-dimensional terminal rectangle. Therefore Q_k was in its list. None of the new facet coordinates is an integer. Hence each refined integer rectangle is entirely inside a listed S_k or disjoint from every listed S_k. An unlisted cluster cannot contain one of its integer points, by the same closure argument. Strict disjointness then implies that each integer rectangle is inside one cluster or outside every cluster.

In particular, with the explicit dimension-only constant

$$A_D = 2^{D+1}(2^D - 1)(2^{2D} + 1)^D, \qquad |\mathcal{P}| \le A_D \max\{1, J\}, \tag{8}$$

we have derived the required partition. The constant is deliberately loose and no sharpness or algorithm running-time claim is made. For $n = 1$ and for singleton input cubes all constructions remain valid.

2.6 The actual coefficient and fixed-penalty consequence

Suppose, in addition, that for fixed $c > 0$ and $\beta \ge 1$ the actual positive scalar starting coefficients obey

$$\begin{aligned} a(x) &\ge cL_k(L_k - 1)/4 \quad (x \in S_k), \\ a(x) &\ge cn(n-1)/4 \quad (x \notin \cup_k S_k), \\ a(x) &\le \beta L_k^2 \quad (x \in S_k). \end{aligned}$$

These coefficient/cube assumptions are retained, not inferred from the critical moment. On an interval of l integer vertices, pairwise variance, path Cauchy–Schwarz and at most $l^2/4$ pairs through one cut give

$$\sum |u - \bar{u}|^2 \le \frac{l(l-1)}{4} \sum_{\text{edges}} |du|^2.$$

For completeness, product variance tensorization follows from the commuting coordinate averages $P_i u = l_i^{-1} \sum_{t=0}^{l_i - 1} u(x_1, \dots, t, \dots, x_D)$: the telescoping orthogonal decomposition of $I - \prod_i P_i$ gives a sum bounded by $\sum_i \|u - P_i u\|_2^2$. Applying the interval bound on its fibers proves

$$\mathrm{Var}_R(u) \le \frac{L(R)(L(R)-1)}{4} \sum_{e \subset R} |du(e)|^2.$$

Inside a cluster $L(R) \le L_k$, and outside all clusters $L(R) \le n$. The displayed coefficient lower bounds therefore give $c\,\mathrm{Var}_R(u) \le \mathcal{E}_{a,R}(u)$. Singleton residuals are zero, so no zero lower bound is divided by. Sum over all field/internal coordinates and all partition cells. Original internal positive-axis edge occurrences are disjoint; omitted inter-cell and periodic-wrap occurrences have nonnegative energy. Thus the ACTUAL normalized means

$$(C_{\mathcal{P}}U)_{R,j} = \sqrt{|R|}\,\mathrm{mean}_R\, U_j$$

satisfy

$$c\|U\|^2 \le \mathcal{E}_a(U) + c\|C_{\mathcal{P}}U\|^2$$

for every cochain, with output dimension $h|\mathcal{P}|$, $h = D(N_c^2 - 1)$. This uses an all-cochain variance estimate, not the invalid inference from a kernel-only margin to fixed penalty c. Means need not preserve the divergence constraint. The inequality consequently also holds on that constraint subspace without changing $C_{\mathcal{P}}$.

If $p = D/2$, $\mu \ge 1$, and the original critical mass bound is $\sum_X a^{-p} \le Mn^D/\mu^p$, then each supplied cube contributes at least $L_k^D/(\beta L_k^2)^p = \beta^{-p}$ to this mass. Hence $J \le \beta^p Mn^D/\mu^p$, and (8) gives

$$\dim F \le hA_D(1 + 2^p\beta^p M)\max\{1, n^D/(1+\mu)^p\}.$$

This proves the restricted supplied-cube consequence with an explicit constant and without a fat-object BSP theorem as a black box. It does not construct the supplied cubes from the critical moment, remove either scale bound or the strong complement bound, resolve the unrestricted critical rectangular question, or establish a covariant/continuum gap. No new originality or independent-discovery claim is made: the compressed splitting strategy is attributed classical geometry, here derived and adapted completely for this special class. The frozen reviewed PDF and its full reports and pending permission remain unchanged.

3 Finite reproduction evidence and its limits

The construction was implemented using exact rational arithmetic. On 5 November 2026 a newly created, exclusively owned Colab Linux CPU runtime with high RAM ran source commit

```
3c309786dff766a97638b9bb4f654bcb14cb9455
```

under Python 3.13.18. Normal and optimized Python each accepted eight baseline certificates and rejected all 47 saved malformed certificates and two invalid inputs. All 58 canonical JSON files matched byte for byte between the two modes. The source had zero assertion nodes; acceptance used explicit failure checks. Both subprocess exit codes were zero. Seven scenes used exhaustive lattice coverage. The scene with root side 10^9 used exact volume equality and pairwise disjointness instead. The checker reconstructed split geometry, active corner sets and potential decrease, and checked whole-rectangle membership and coverage. Midpoint and corner-shrink events both occurred. Full logs, real exit sentinels, actual malformed certificates and sources are retained in the archive.

This checker shares primitives and authorship with the builder. It is not an independently authored validator or reproduction in two independent fresh clones. These are finite fixture results, not a proof certificate for the universal theorem or Lean verification. All baseline first leaves had proper occupancy at most one. They therefore did not test the multiple-object family proved in the preceding part. Closed-contact negative controls in dimensions two, three and four had respectively five, nine and seventeen closed contacts but no proper contact.

The archive is 193158 bytes. After recovery from Colab all 123 inner manifest members were rehashed without a mismatch. Its SHA256 is

```
64f548dff043cf6cf9f1c1a3abf24fc79
3cf6565ff7dd436c1d0f263351b5be1
```

The line break in this printed hash is typographical; concatenate the two lines. The full archive is included in the reproduction package. The runtime was disconnected after recovery. The driver took approximately 3.88 seconds; this is not the billed connection time or an algorithmic complexity bound. The resource panel displayed 50.99 GB RAM; an exact billed compute-unit delta was unavailable and is not inferred.

4 Model assessments and review context

This exact companion PDF has no external model assessment and no peer review. Preparing-assistant analytic checking and visual layout checking are reported separately from external review. The authoring interface is OpenAI Codex; no more specific API-attested model identifier is available in this record. No independence from the proof's preparing context is claimed. No review score is assigned to this artifact.

For historical context only, the distinct frozen manuscript identified by the SHA256 prefix `eb4df3` has its own complete first model report and review context retained in revision136. The original report gives 3.5/10 and minor revision; the raw model identifier is unknown and the operator observed Google Gemini 4 Argon High in the interface. Its five material objections concern uninspected formal/Colab evidence, the non-Lean analytic continuation, primary sources not inspected by that reviewer, the still-open unrestricted rectangular question, and the absent interacting/continuum connection. These remain part of that artifact's report. They have not been erased or reassessed by this companion. That review was not blinded; prior assessment narrative was in its PDF. Independence and context isolation were not established.

The public AIRR v5 is a separate artifact, a working paper rather than an accepted or peer-reviewed paper. Its own first report and operator response remain separate. Neither

historical rating transfers to this new companion. Public deposit, model evaluation, editorial acceptance and peer review are different events; no new such event is claimed here.

5 Unresolved obligations and reproducibility instructions

The unrestricted critical rectangular partition question is open. The generic common-refinement mechanism remains stopped after its recorded counterexamples; these classical proof completions do not restart it. All actual cube, coefficient lower/upper, strong complement and moment assumptions in part two remain. No interacting physical hypothesis is removed. PR29 remains recorded as 20/41 with TermSource zero.

For reproduction use the manifest's source SHA, compile this main file with the two pinned fragments, and retain every LaTeX pass, the exact PDF hash and all page images. The render driver runs only in this chat's own Colab Linux runtime. The unit139 fixtures are already preserved; rendering this companion does not rerun them. The master and fragments are not an alteration of the frozen reviewed manuscript, and their successful compilation would not establish the mathematics.